LD0-B Effective Density-Broadening Specification

CIC-phase covariance, canonical-stencil covariance, and versioned adaptive resolution

mdstats development specification

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Contents

1 Status and scope

This specification implements architecture gate **LD0-B** for `mdstats 0.19.44a0`. It is subordinate to the *Dynamical Framework and Atomic Density Architecture Standard*.

The gate introduces the explicit broadening metric `effective_cic_stencil_rms_v1`. It combines the covariance of periodic trilinear cloud-in-cell assignment with the covariance of the canonical `discrete_periodized_v1` smoothing stencil.

The existing `gaussian_sigma_v1` metric remains the default. No existing call changes its resolved grid or density values unless the new metric is selected explicitly.

This gate does not implement:

- local-sparse storage or sparse convolution;
- automatic dense/sparse backend selection;
- framework-edge quadrature refinement;
- probability-shell or mesh changes;
- sample-dependent Gaussian bandwidths.

2 Scientific ownership

The trilinear cloud-in-cell assignment follows the particle-mesh construction of Hockney and Eastwood (1988). Covariance addition for centered independent smoothing kernels is standard probability theory. The phase-resolved periodic CIC covariance, its combination with the canonical stencil, the deterministic refinement search, and the migration policy are project-specific derivations and policies.

3 Logical sampling basis

Let the row-vector display cell be

$$H = \begin{bmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \mathbf{a}_3 \end{bmatrix},$$

and let the logical grid shape be

$$\mathbf{N} = (N_1, N_2, N_3).$$

The real-space sampling-basis rows are

$$\mathbf{b}_a = \frac{\mathbf{a}_a}{N_a}.$$

For a folded fractional sample $\mathbf{f}_s$, define the grid phase

$$\mathbf{d}_s = \mathbf{N} \odot \mathbf{f}_s - \lfloor \mathbf{N} \odot \mathbf{f}_s \rfloor, \qquad 0 \le d_{sa} < 1.$$

The phase computation uses exactly the same logical-node convention as periodic CIC deposition.

4 CIC assignment covariance

Along axis a, CIC assigns the sample to the lower node with probability $1 - d_{sa}$ and to the upper node with probability d_{sa}. Relative to the sample, the two displacements are

$$-d_{sa}\mathbf{b}_a \quad \text{and} \quad (1 - d_{sa})\mathbf{b}_a.$$

Their weighted mean is zero and their covariance contribution is

$$C_{\mathrm{CIC},s,a} = d_{sa}(1 - d_{sa})\mathbf{b}_a\mathbf{b}_a^T.$$

Thus

$$C_{\mathrm{CIC},s} = \sum_{a=1}^{3} d_{sa}(1 - d_{sa})\mathbf{b}_a\mathbf{b}_a^T.$$

For nonnegative source weights w_s with positive total weight, define

$$\omega_s = \frac{w_s}{\sum_t w_t},$$

and

$$\overline{C}_{\mathrm{CIC}} = \sum_s \omega_s C_{\mathrm{CIC},s}.$$

Atomic and framework-vertex occupancy use repeated frame weights. Framework-edge quadrature uses its arc-length quadrature weights when reporting the realized edge field diagnostic. When framework-edge resolution is inherited from framework vertices, metadata records `resolution_reference_source="framework_vertices"`.

5 Canonical-stencil covariance

For the unaggregated retained image contributions $a_{\mathbf{q}}$ defined by `discrete_periodized_v1`, let

$$\Delta\mathbf{x}_{\mathbf{q}} = \left(\frac{q_1}{N_1}, \frac{q_2}{N_2}, \frac{q_3}{N_3}\right) H.$$

The normalized canonical-stencil covariance is

$$C_g = \frac{\sum_{\mathbf{q}} a_{\mathbf{q}} \Delta\mathbf{x}_{\mathbf{q}} \Delta\mathbf{x}_{\mathbf{q}}^T}{\sum_{\mathbf{q}} a_{\mathbf{q}}}.$$

It is evaluated by the same bounded support enumeration used to build the canonical stencil, but a covariance-only path must not allocate the dense stencil array.

For $\sigma = 0$,

$$C_g = 0.$$

6 Effective artificial broadening

The total artificial covariance is

$$C_{\mathrm{art}} = \overline{C}_{\mathrm{CIC}} + C_g.$$

The scalar broadening metric is the isotropic RMS component width

$$s_{\mathrm{art}} = \sqrt{\frac{\mathrm{tr}(C_{\mathrm{art}})}{3}}.$$

The component diagnostics are

$$s_{\mathrm{CIC}} = \sqrt{\frac{\mathrm{tr}(\overline{C}_{\mathrm{CIC}})}{3}}, \qquad s_g = \sqrt{\frac{\mathrm{tr}(C_g)}{3}}.$$

The adaptive target is

$$s_{\mathrm{art}} \le \alpha s_q,$$

where s_q is the approved positional-spread reference and the default is $\alpha = 0.5$.

7 Compatibility and selection rules

The valid implemented combinations are:

Broadening metric	Smoothing operator	Status
`gaussian_sigma_v1`	`legacy_spectral_v1`	supported and default
`gaussian_sigma_v1`	`discrete_periodized_v1`	supported
`effective_cic_stencil_rms_v1`	`discrete_periodized_v1`	supported explicitly
`effective_cic_stencil_rms_v1`	`legacy_spectral_v1`	rejected

The effective metric is not a diagnostic for the legacy spectral operator because its stencil covariance is defined by `discrete_periodized_v1`.

Explicit `grid_shape` and explicit `gaussian_bandwidth` remain authoritative. If their resolved effective width exceeds the target, the implementation records the failure and emits a warning; it does not change the explicit value.

8 Automatic resolution search

The effective-width search applies only when:

1. `adaptive_smearing=True`;
2. the broadening metric is `effective_cic_stencil_rms_v1`;
3. no explicit `grid_shape` is supplied;
4. no explicit `gaussian_bandwidth` is supplied;
5. the positional-spread target is finite and valid.

Starting from the nominal interval, the implementation evaluates the exact effective width for the resolved logical shape. On failure, it proposes a smaller interval from the local scale estimate

$$h_{n+1} = h_n \min\left(0.9, 0.98\frac{\alpha s_q}{s_{\mathrm{art},n}}\right).$$

If this proposal leaves the logical shape unchanged, the next exact shape-transition interval is used. The process continues until a passing shape is found or the dense voxel budget is reached. A deterministic bracket refinement then seeks a coarser passing interval while retaining one certified passing shape.

Every accepted automatic result is re-evaluated and must satisfy the target within

$$\tau_s = 5 \times 10^{-13} \max(1, \alpha s_q).$$

If the finest shape allowed by the dense budget fails, the result is marked `adaptive_smearing_budget_limited=True` and the unresolved ratio is recorded.

9 Zero-spread and zero-bandwidth policies

If $s_q = 0$, no positive finite target is defined:

```
adaptive_target_defined = False
adaptive_refinement_applied = False
nominal or explicit resolution retained
warning and metadata emitted
```

For $\sigma = 0$, the stencil covariance is zero but CIC covariance remains. Therefore

$$s_{\mathrm{art}} = s_{\mathrm{CIC}}.$$

An explicit zero bandwidth is preserved. The implementation reports whether the CIC width alone exceeds the positional target.

10 Data structures

10.1 ArtificialBroadeningDiagnostic

```python
@dataclass(frozen=True, slots=True)
class ArtificialBroadeningDiagnostic:
    grid_shape: tuple[int, int, int]
    sample_count: int
    source_weight_sum: float
    phase_variance_coefficients: NDArray[np.float64]  # shape (3,)
    cic_covariance: NDArray[np.float64]                # shape (3, 3)
    stencil_covariance: NDArray[np.float64]            # shape (3, 3)
    total_covariance: NDArray[np.float64]              # shape (3, 3)
    cic_rms: float
    stencil_rms: float
    effective_rms: float
    metadata: FrozenJSONMapping
```

All arrays are defensive, C-contiguous, finite, and read-only.

10.2 PeriodicGaussianStencilMoments

```python
@dataclass(frozen=True, slots=True)
class PeriodicGaussianStencilMoments:
    grid_shape: tuple[int, int, int]
    display_cell: NDArray[np.float64]
    gaussian_bandwidth: float
    kernel_tail_tolerance: float
    cutoff_radius: float
    pre_normalization_sum: float
    normalization_factor: float
    periodic_image_contribution_count: int
    covariance: NDArray[np.float64]
    metadata: FrozenJSONMapping
```

This record uses the canonical support enumerator without allocating a dense logical stencil.

11 Metadata

Fields using the effective metric record:

```
broadening_metric
effective_artificial_rms
cic_assignment_rms
canonical_stencil_rms
cic_covariance_cartesian
stencil_covariance_cartesian
artificial_covariance_cartesian
cic_phase_variance_coefficients
broadening_sample_count
broadening_source_weight_sum
adaptive_target_width
adaptive_target_achieved
```

```
adaptive_target_ratio
resolution_reference_source
```

Atomic and framework-vertex fields use their own samples as the resolution reference. Framework-edge fields additionally report their own realized edge-sample artificial covariance while retaining `resolution_reference_source="framework_v`

12 Public API

The existing option record is used:

```
DensityResolutionOptions(
    broadening_metric="effective_cic_stencil_rms_v1",
)

DensityKernelOptions(
    smoothing_operator="discrete_periodized_v1",
    kernel_tail_tolerance=1.0e-8,
)
```

No default changes in LD0-B.

13 Validation

LD0-B passes only if:

1. analytic CIC covariance matches brute-force eight-node assignment covariance with relative Frobenius error $\leq 5 \times 10^{-13}$;
2. covariance-only canonical-stencil moments match the full stencil covariance with relative Frobenius error $\leq 5 \times 10^{-13}$;
3. on-node CIC phase produces exactly zero CIC covariance;
4. half-node phase produces the analytic maximum $\frac{1}{4}\sum_a \mathbf{b}_a \mathbf{b}_a^T$;
5. weighted atomic, framework-vertex, and edge-quadrature diagnostics conserve source weighting semantics;
6. every automatically accepted finite target satisfies the declared effective-width inequality;
7. budget-limited, explicit-grid, explicit-bandwidth, zero-spread, and zero-bandwidth cases follow the declared warning and metadata policies;
8. `gaussian_sigma_v1` plus `legacy_spectral_v1` remains byte-for-byte compatible with `mdstats 0.19.43a0`;
9. the default broadening metric and default smoothing operator do not change.

14 Failure policy

The implementation fails before density allocation when:

- the effective metric is paired with `legacy_spectral_v1`;
- sample positions or weights are invalid;
- source weights are negative or have nonpositive total weight;
- canonical support enumeration exceeds its declared complexity limit;
- a resolved automatic shape exceeds the field voxel budget.

An unresolved target caused by an explicit input or finite resource budget is reported honestly and is not converted into an exception unless the existing resource limit is itself exceeded.

15 References

1. Hockney, R. W., and J. W. Eastwood. *Computer Simulation Using Particles.* Taylor & Francis, 1988. CIC assignment is adapted from this source.
2. The covariance addition and Gaussian covariance identities are standard probability background. The periodic phase-resolved combination and adaptive selection policy are project-specific.